

LECTURE 1

From the Naturals to the Reals: Induction, Ordered Fields, and Completeness

MATH S4061 | Introduction to Modern Analysis I

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God made the integers; all else is the work of man.

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Overview. The course opens with a deceptively simple question—is $0.999\dots = 1$?—whose honest answer requires building the real numbers from the ground up. We begin with the natural numbers and induction, construct $\mathbb{Z}$ and $\mathbb{Q}$, and isolate the field and order axioms they satisfy. The rationals then fall short for analysis: $\sqrt{2}$ is missing, and the sets bounding it have neither a largest nor a smallest element. Repairing that gap is the *least-upper-bound property*, which characterises $\mathbb{R}$; from it follow the Archimedean property and the density of $\mathbb{Q}$ —and, at last, the resolution of $0.999\dots = 1$.

1.1 The natural numbers and induction

Definition 1.1.1

The natural numbers

The *natural numbers* are $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ ¹, generated from 0 by the *successor function*

$$S : \mathbb{N} \rightarrow \mathbb{N}, \quad n \mapsto n + 1.$$

Axiom 1.1.2

Induction

If $A \subseteq \mathbb{N}$ satisfies $0 \in A$ and $n \in A \Rightarrow n + 1 \in A$, then $A = \mathbb{N}$.

Theorem 1.1.3

The principle of mathematical induction

Let $P(n)$ be a statement for each $n \in \mathbb{N}$. If $P(0)$ holds, and $P(n) \Rightarrow P(n + 1)$ for every n , then $P(n)$ holds for all $n \in \mathbb{N}$.

Proof 1.1.3

Proof of the Induction Principle from the Axiom.

- (1) Let $A = \{n \in \mathbb{N} : P(n) \text{ holds}\}$.
- (2) Since $P(0)$ holds, $0 \in A$; and $P(n) \Rightarrow P(n + 1)$ says $n \in A \Rightarrow n + 1 \in A$.
- (3) By the induction axiom (1.1.2), $A = \mathbb{N}$; that is, $P(n)$ holds for every n .

[Direct] uses 1.1.2. ■

Example 1.1.4

Gauss's sum

For every integer $n \geq 1$, $1 + 2 + \dots + n = \frac{n(n+1)}{2}$.

Proof 1.1.4

Proof by Induction of Gauss's Formula.

- (1) For $n = 1$ the formula reads $1 = \frac{1 \cdot 2}{2}$, which holds.
- (2) Assume $1 + \dots + n = \frac{n(n+1)}{2}$. Adding $n + 1$ to both sides,

$$1 + \dots + n + (n + 1) = \frac{n(n+1)}{2} + (n + 1) = \frac{(n+1)(n+2)}{2},$$

which is the formula for $n + 1$. The result follows by induction (1.1.3). ■

¹We take $0 \in \mathbb{N}$; Rudin instead starts the positive integers at 1. The convention is harmless—only the base of an induction shifts (Gauss's sum, 1.1.4, begins at $n = 1$).

[A10] from the base $n = 1$ (apply 1.1.3 to the shifted statement).

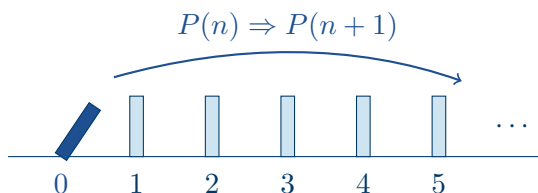


Figure 1.1.5. INDUCTION AS TOPPLING DOMINOES. The base case $P(0)$ topples the first domino; the inductive step $P(n) \Rightarrow P(n+1)$ passes the fall to the next, so every $P(n)$ falls in turn—the conclusion of 1.1.2 and 1.1.3.

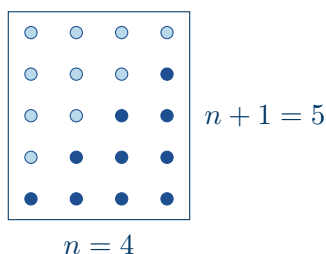


Figure 1.1.6. GAUSS'S SUM AS HALF A RECTANGLE. Two copies of the staircase $1 + 2 + 3 + 4$ interlock into a 4×5 grid, so $2(1 + 2 + 3 + 4) = 4 \cdot 5$; in general $1 + \cdots + n = \frac{n(n+1)}{2}$ (1.1.4).

1.2 The integers and the rationals

Construction 1.2.1

The integers

The *integers* are signed naturals with the two zeros glued together:

$$\mathbb{Z} := (\{+, -\} \times \mathbb{N}) / \sim, \quad (+, 0) \sim (-, 0).$$

Writing $+n, -n$ for the classes recovers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$.

Construction 1.2.2

The rationals

The *rational*s are equivalence classes of integer pairs with nonzero second coordinate:

$$\mathbb{Q} := \{(p, q) \in \mathbb{Z} \times \mathbb{Z} : q \neq 0\} / \sim, \quad (p, q) \sim (r, s) \iff ps = rq.$$

The class of (p, q) is written $\frac{p}{q}$.

Remark 1.2.3

One number, many fractions

A rational *is* the whole class, e.g. $[\frac{1}{2}] = \{\frac{1}{2}, \frac{2}{4}, \frac{-1}{-2}, \frac{17}{34}, \dots\}$. In practice one names it by its *preferred representative*: the irreducible fraction $\frac{a}{b}$ with $\gcd(a, b) = 1$ and the sign carried in the numerator.

1.3 Fields and ordered fields

Definition 1.3.1

Field

A *field* is a set F with two operations $+: F \times F \rightarrow F$ and $\cdot: F \times F \rightarrow F$ satisfying eleven axioms²—making F an abelian group under $+$, making $F \setminus \{0\}$ an abelian group under $\cdot$, with multiplication distributing over addition.

Definition 1.3.2

Order

An *order* on a set S is a relation $<$ such that

- (1) (*trichotomy*) for $x, y \in S$, exactly one of $x < y$, $y < x$, $x = y$ holds;
- (2) (*transitivity*) $x < y$ and $y < z$ imply $x < z$.

A set carrying an order is *totally* (or *linearly*) ordered.

Definition 1.3.3

Ordered field

An *ordered field* is a field F with an order $<$ compatible with the operations:

- (1) $y < z \Rightarrow x + y < x + z$ for all x ;
- (2) $x > 0$ and $y > 0 \Rightarrow xy > 0$.

Remark 1.3.4

The rationals are an ordered field

$\mathbb{Q}$ is an ordered field. The rest of the lecture asks what it *lacks*—and why that makes it unfit for analysis.

1.4 Why the rationals are not enough

Proposition 1.4.1

No rational squares to 2

There is no $x \in \mathbb{Q}$ with $x^2 = 2$.

Proof 1.4.1

Proof by Contradiction of the Irrationality of $\sqrt{2}$.

²Five for addition (A1–A5), five for multiplication (M1–M5), and the distributive law (D); the full statements are R 1.12.

- (1) Suppose $x = \frac{p}{q} \in \mathbb{Q}$ is irreducible with $x^2 = 2$. Then $p^2 = 2q^2$, so p^2 is even, hence p is even (an odd integer has an odd square); write $p = 2k$.
- (2) Substituting, $4k^2 = 2q^2$, so $q^2 = 2k^2$ is even, hence q is even.
- (3) Then $2 \mid \gcd(p, q)$, contradicting that $\frac{p}{q}$ is irreducible. So no such x exists. ■

[A9] uses “ n^2 even $\Rightarrow n$ even”.

Proposition 1.4.2

The rationals have a gap

Let $A = \{p \in \mathbb{Q} : p > 0, p^2 < 2\}$ and $B = \{p \in \mathbb{Q} : p > 0, p^2 > 2\}$. Then A has no largest element and B has no smallest element.

Proof 1.4.2

Proof by an Explicit Construction of the Gap.

- (1) Given $p > 0$, set

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > 0.$$
- (2) A computation gives

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2},$$
 so $q^2 - 2$ has the same sign as $p^2 - 2$: if $p \in A$ then $q \in A$, and if $p \in B$ then $q \in B$.
- (3) Also $q - p = -\frac{p^2 - 2}{p + 2}$, so $q > p$ when $p \in A$ and $q < p$ when $p \in B$.
- (4) Hence every $p \in A$ is exceeded by a larger $q \in A$, so A has no largest element; and every $p \in B$ is undercut by a smaller $q \in B$, so B has no smallest element. ■

[Constr] cf. R 1.1.

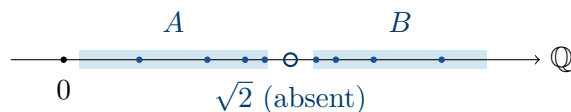


Figure 1.4.3. THE GAP AT $\sqrt{2}$. The positive rationals A ($p^2 < 2$) climb toward $\sqrt{2}$ with no largest member; B ($p^2 > 2$) descend toward it with no smallest; in $\mathbb{Q}$ the meeting point $\sqrt{2}$ is simply absent.

1.5 Bounds and the supremum

Definition 1.5.1

Upper bound

Let S be an ordered set and $E \subseteq S$. An element $\beta \in S$ is an *upper bound* for E if $x \leq \beta$ for every $x \in E$.

Definition 1.5.2

Least upper bound (supremum)

β is the *least upper bound* of E if it is an upper bound for E and no smaller element is: $\alpha < \beta$ implies α is not an upper bound. We write $\beta = \sup E = \text{lub } E$.

Definition 1.5.3

Lower bound and infimum

Dually, $\beta \in S$ is a *lower bound* for E if $\beta \leq x$ for every $x \in E$, and the *greatest lower bound* (or *infimum*) if no larger element is a lower bound: $\beta < \alpha$ implies α is not a lower bound. We write $\beta = \inf E = \text{glb } E$. Since $\inf E = -\sup(-E)$, the least-upper-bound property (1.6.1) carries a matching greatest-lower-bound property.

1.6 The real number system

Theorem 1.6.1

The real number system

There is an ordered field $\mathbb{R} \supseteq \mathbb{Q}$ in which every nonempty subset that is bounded above has a least upper bound. It is unique up to isomorphism.³

Remark 1.6.2

Completeness and density

The property in 1.6.1 is *completeness*: $\mathbb{R}$ is complete iff it has the least-upper-bound property. A first consequence is that $\mathbb{Q}$ is *dense* in $\mathbb{R}$ —between any two reals lies a rational (1.7.2).

Remark 1.6.3

Three ways to build $\mathbb{R}$

- *Accept it*—take 1.6.1 on faith and learn to use sup.
- *Dedekind cuts* (1.6.4)—realise each real as a downward-closed set of rationals.
- *Completion*—the general procedure that completes any metric space (deferred).

Definition 1.6.4

Dedekind cut

A *cut* is a set $\alpha \subseteq \mathbb{Q}$ such that

- (1) $\alpha \neq \emptyset$ and $\alpha \neq \mathbb{Q}$;
- (2) α is downward closed: $r \in \alpha$ and $s < r$ imply $s \in \alpha$;
- (3) α has no largest element.

The cuts, suitably ordered, *are* the real numbers.

³An isomorphism of ordered fields is a bijection preserving $+$, $\cdot$, and $<$; “unique up to isomorphism” means any two complete ordered fields are indistinguishable as such. Two were built in 1872: Dedekind’s cuts (*Stetigkeit und irrationale Zahlen*) and Cantor’s equivalence classes of Cauchy sequences of rationals.

Example 1.6.5 $\sqrt{2}$ as a cutThe real number $\sqrt{2}$ is the cut

$$\alpha = \{q \in \mathbb{Q} : q < 0\} \cup \{q \in \mathbb{Q} : q^2 < 2\}.$$

This is the set A of 1.4.2 together with the non-positive rationals; the former “gap” is now a bona fide element.

1.7 The Archimedean property, density, and decimals

Theorem 1.7.1

Archimedean property

If $x, y \in \mathbb{R}$ and $x > 0$, then there is a positive integer n with $nx > y$.⁴

Proof 1.7.1

Proof by Contradiction of the Archimedean Property.

- (1) Suppose not: the set $A = \{nx : n \in \mathbb{N}\}$ is bounded above. By the least-upper-bound property (1.6.1) it has a supremum $\alpha = \sup A$.
- (2) Since $x > 0$, $\alpha - x < \alpha$, so $\alpha - x$ is not an upper bound of A : some $m \in \mathbb{N}$ has $mx > \alpha - x$.
- (3) Then $(m+1)x > \alpha$, yet $(m+1)x \in A$ —contradicting $\alpha = \sup A$. So A is not bounded above, and in particular some $nx > y$.

■

[A5+A9] via the supremum; uses 1.6.1.

Theorem 1.7.2Density of $\mathbb{Q}$ in $\mathbb{R}$

If $x, y \in \mathbb{R}$ and $x < y$, then there is a rational q with $x < q < y$.

Proof 1.7.2

Proof by the Archimedean Property of the Density of $\mathbb{Q}$.

- (1) Since $y - x > 0$, the Archimedean property (1.7.1) gives $n \in \mathbb{N}$ with $n(y - x) > 1$, i.e. $ny - nx > 1$.
- (2) Since $ny - nx > 1$, the interval (nx, ny) contains an integer m (§1.A.1).
- (3) Dividing by $n > 0$, $x < \frac{m}{n} < y$, so $q = \frac{m}{n}$ works.

■

[Direct] uses 1.7.1.

⁴Named by Otto Stolz in the 1880s after Archimedes, who postulated it in *On the Sphere and the Cylinder*; the idea is older still—Euclid records it (*Elements* V, Def. 4) and credits Eudoxus.

Construction 1.7.3

Real numbers as decimals

Every *nonnegative* real is the supremum of its finite decimal truncations (a negative real is then named by its sign):

$$1.4112\dots \mapsto \sup\{1, 1.4, 1.41, 1.411, \dots\}.$$

Example 1.7.4

$$0.999\dots = 1$$

This settles the opening question. Put $x = 1 - 0.999\dots$; then $0 \leq x < 10^{-k}$ for every $k \in \mathbb{N}$. By the Archimedean property (1.7.1)⁵ no positive real lies below every 10^{-k} , so $x = 0$ —that is, $0.999\dots = 1$. More generally every terminating decimal equals its repeating-9 twin ($2.5 = 2.4999\dots$, and so on), so

$$\mathbb{R} = \{\text{decimal expansions}\} / \sim,$$

the quotient that identifies each terminating decimal with that twin.

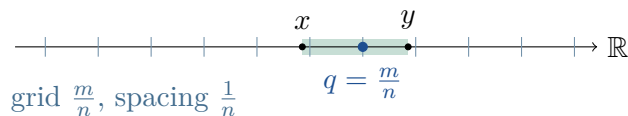


Figure 1.7.5. DENSITY OF $\mathbb{Q}$ IN $\mathbb{R}$. Choosing n with $1/n < y - x$ makes the grid of multiples m/n finer than the gap, so some m/n lands strictly between x and y (1.7.2).

⁵If $x > 0$, the Archimedean property gives n with $nx > 1$; taking k with $10^k \geq n$ then yields $10^k x > 1$, i.e. $x > 10^{-k}$.

Appendix

Supplementary material—additional proofs, context, and other treatments; not part of the lecture proper.

Supplement 1.A.1

An interval longer than 1 contains an integer

If $s, t \in \mathbb{R}$ with $t - s > 1$, then there is an integer m with $s < m < t$. (Used silently in the density proof, 1.7.2.)

Proof 1.A.1

Direct Proof from the Archimedean Property.

- (1) The integers exceeding s are nonempty (by the Archimedean property) and bounded below by s , so they have a least element m ; thus $m - 1 \leq s < m$.
- (2) Hence $m \leq s + 1 < t$, and therefore $s < m < t$.

■

[Direct] uses 1.7.1; cf. R 1.20.

Supplement 1.A.2

Every nonnegative real has a decimal expansion

Construction 1.7.3 names each nonnegative real as the supremum of its truncations; here we build those truncations digit by digit and confirm the supremum is the number we began with. Given $x \geq 0$, there are an integer $a_0 \geq 0$ and digits $d_1, d_2, \dots \in \{0, \dots, 9\}$ whose truncations

$$t_k = a_0 + \sum_{i=1}^k \frac{d_i}{10^i} \quad \text{satisfy} \quad 0 \leq x - t_k < 10^{-k},$$

so that $x = \sup_k t_k$. The rule below always takes the *largest* admissible digit, so a number with a terminating expansion receives it ($1 = 1.000\dots$), and the repeating-9 twin (1.7.4) is the single expansion this rule never produces.

Proof 1.A.2

Direct Proof by Greedy Digit Selection.

- (1) For the integer part, let a_0 be the greatest integer with $a_0 \leq x$ — it exists by the least-integer argument of 1.A.1, applied to the integers exceeding x , and $a_0 \geq 0$ since $x \geq 0$. Then $t_0 = a_0$ obeys $0 \leq x - t_0 < 1$.
- (2) For the digits, suppose t_{k-1} has been built with $0 \leq x - t_{k-1} < 10^{-(k-1)}$. Let d_k be the largest digit in $\{0, \dots, 9\}$ with $t_{k-1} + d_k 10^{-k} \leq x$; the digit 0 qualifies, and $d_k \leq 9$ because $x < t_{k-1} + 10 \cdot 10^{-k} = t_{k-1} + 10^{-(k-1)}$. Put $t_k = t_{k-1} + d_k 10^{-k}$. Then $t_k \leq x$, while maximality of d_k gives $x < t_k + 10^{-k}$; hence $0 \leq x - t_k < 10^{-k}$ and the invariant persists.
- (3) For the supremum: every $t_k \leq x$, so x is an upper bound. If $\beta < x$, then $x - \beta > 10^{-k}$ for some k (Archimedean property, 1.7.1), so $t_k > x - 10^{-k} > \beta$; thus no

$\beta < x$ bounds the truncations, and $x = \sup_k t_k$.



[Direct] uses 1.7.1, 1.A.1.

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